Vol 4 Chapter 3 — The BST-SR / BST-GR Boundary

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Chapter 3 — The BST-SR / BST-GR Boundary

Standard physics treats special relativity and general relativity as distinct theoretical frameworks — SR for flat spacetime, GR for curved spacetime, with the curvature switching on whenever matter is present. The transition between the two is structural in standard physics but unmotivated.

In BST, the SR–GR transition is *substrate-mechanical*. SR is the substrate's behavior at scales where the residual Bergman curvature is locally flat; GR is the substrate's behavior when accumulated commitment-cycle curvature becomes significant. The crossover happens at the Koons tick scale combined with the substrate's Casimir-eigenvalue spread.

3.1 SR as substrate at Scale 1

Special relativity emerges at substrate Scale 1 (Volume 0 Chapter 3 §3.4) — within a single Koons-tick commitment cycle. At this scale, the substrate's residual Bergman curvature is approximately flat; the substrate's symmetry group $SO_0(5,2)$ restricts to the Lorentz subgroup $SO_0(3,1)$ acting on the four-dimensional Minkowski conformal boundary; the Poincaré algebra is the substrate's natural symmetry.

This is BST-SR: special relativity as the substrate's intra-cycle behavior at locally flat residual curvature.

3.2 GR as substrate at Scale 3

General relativity emerges at substrate Scale 3 — accumulated commitment cycles at large scales. The residual Bergman curvature integrates across many ticks; at distances above the Bergman crossover scale, the integrated curvature becomes the gravitational metric we observe.

This is BST-GR: general relativity as the substrate's inter-cycle long-distance behavior with integrated residual curvature.

3.3 The crossover scale

The substrate's BST-SR to BST-GR crossover happens at a specific scale set by the Koons tick combined with the Casimir-eigenvalue spread. The substrate's local-flatness regime extends up to scales of order $t_{\text{Planck}}^{1/2} \cdot C_2^{-1/2}$; above this scale, the residual curvature is observable as gravitational effects.

For laboratory experiments, the crossover scale is far below any measurement: gravity is detectable only at macroscopic scales because the substrate's local-flatness regime extends well above all atomic and molecular sizes.

3.4 What comes next

Chapter 4 — the cosmological constant from substrate vacuum — already rewritten as the second Vol 4 recruiter chapter.

Where to look this up: T2106 (gravity as residual substrate curvature). Lorentz invariance from $SO_0(5,2)$ restriction: Volume 0 Chapter 4 §4.5. For standard SR/GR: Carroll, *Spacetime and Geometry*; Misner–Thorne–Wheeler.